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Outline: Week 01 A: Monday, B: Thursday

I) An Odyssey

II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

Outline: Week 02 A: Monday, Course Logistics & Topics

Outline: W01 B

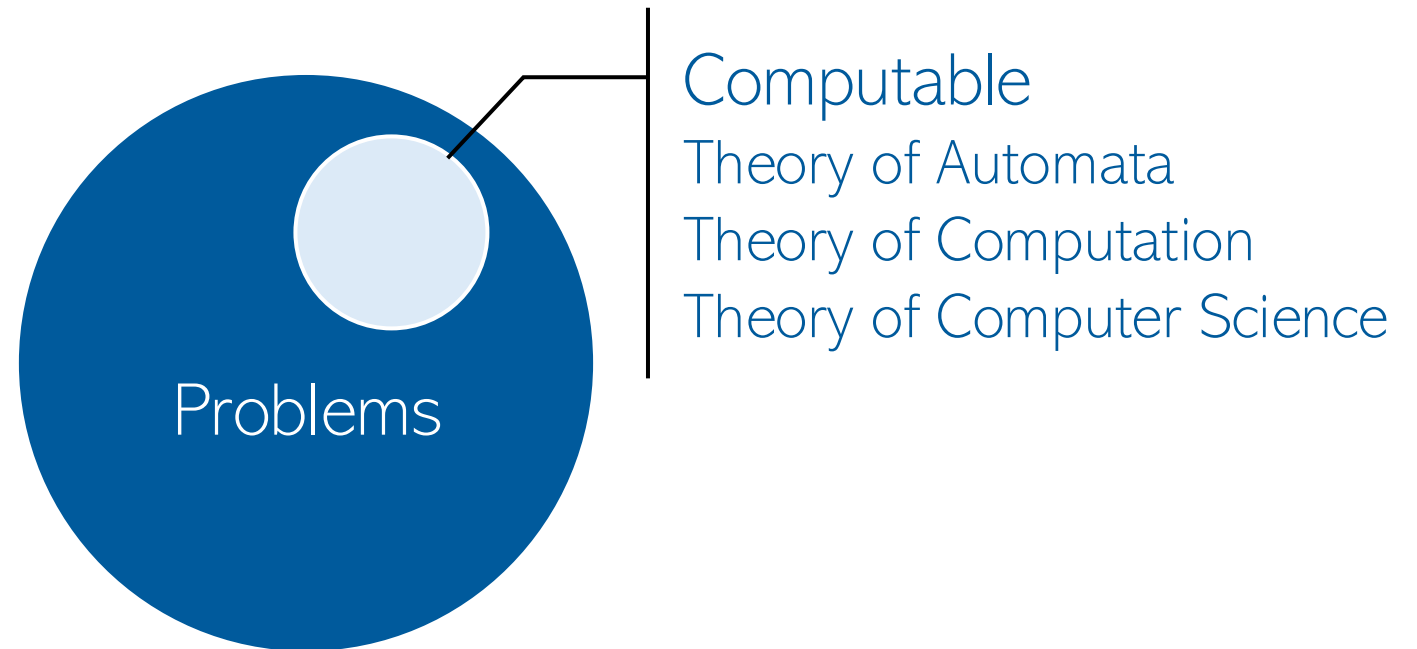
I) An Odyssey

II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

A Data Odyssey



Incompleteness: Not all problems have solutions!

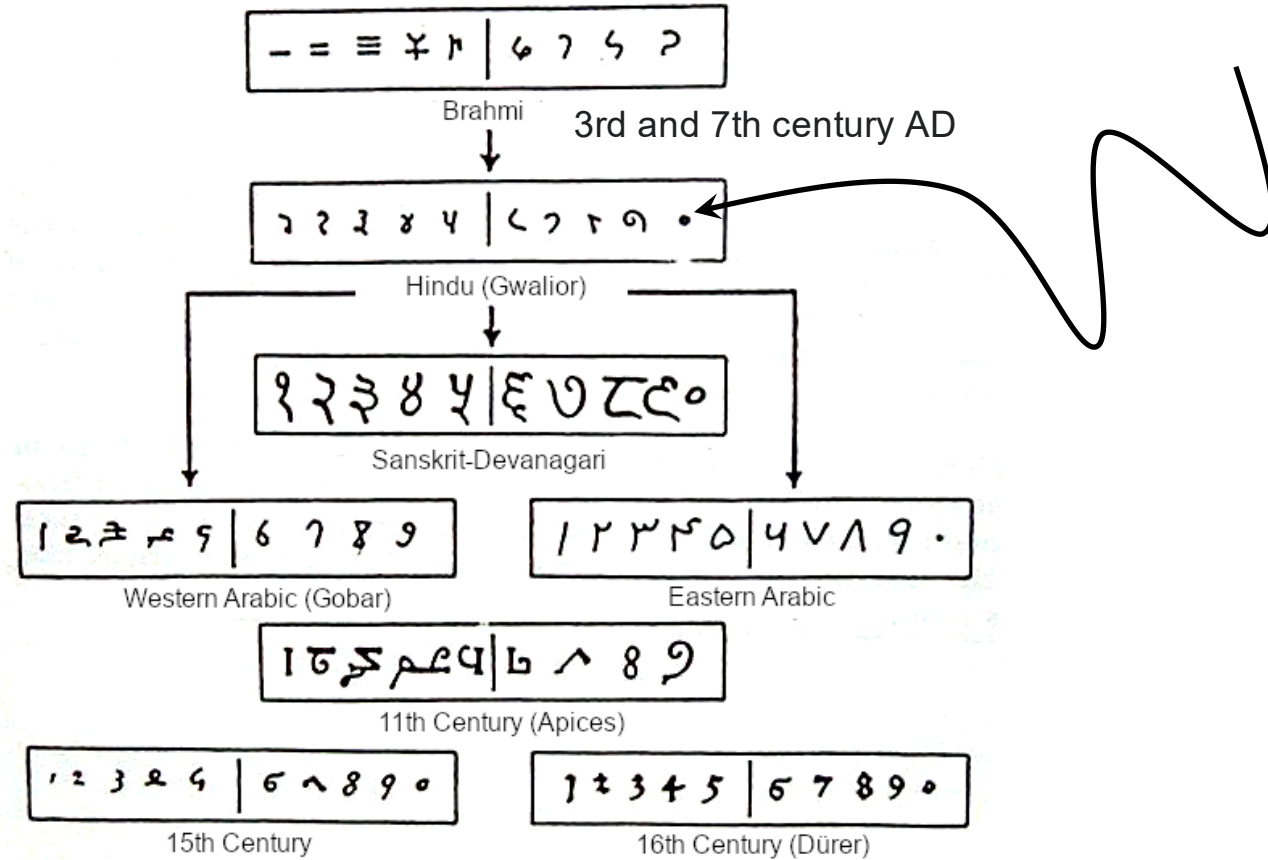
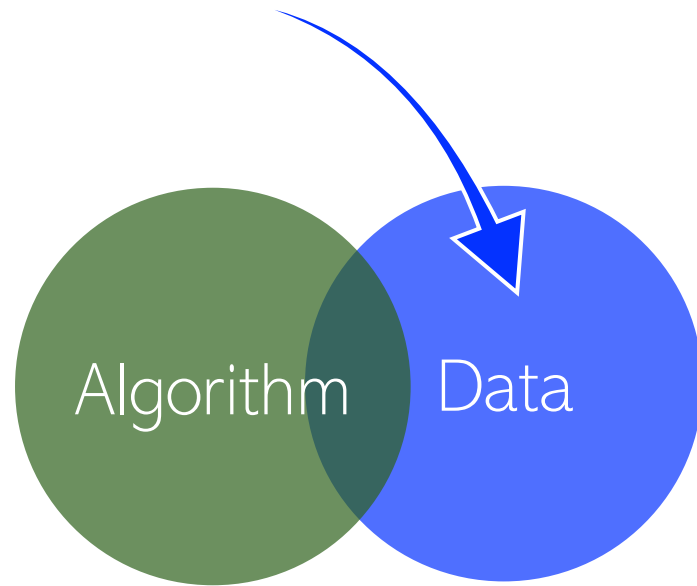
Undecidability: Even if there is a solution, we may not obtain it!

Complexity (Time): Even if we can obtain it, it may take forever (NP)!

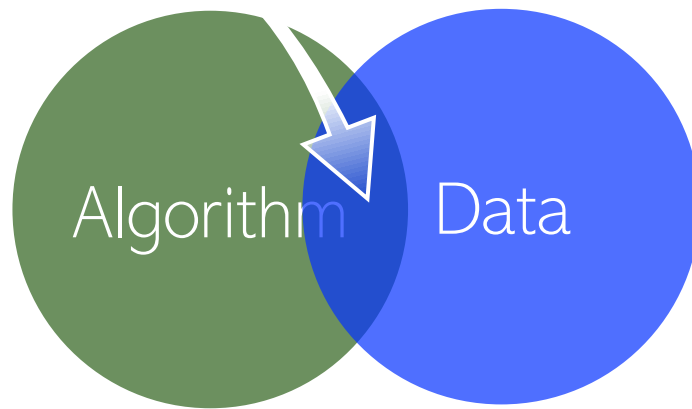
Complexity (Space): Even if we can obtain it, it may need huge space!

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A Data Odyssey



A Data Odyssey



Data gets bigger!
Algorithms get slower!

Data Engineers ↔ Algorithm Designers

Better Data Modeling ↔ Faster Algorithm

- Sorting in Arrays vs. Trees
- Searching in Arrays vs. Trees
- Linear Algebra on Sparse Matrix in Arrays vs. Linked Lists
- ...

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A Data Odyssey

Maya ~750 BC.

A vigesimal (/vɪˈdʒɛsɪməl/) or base-20 numeral system.

https://en.wikipedia.org/wiki/Maya_numerals

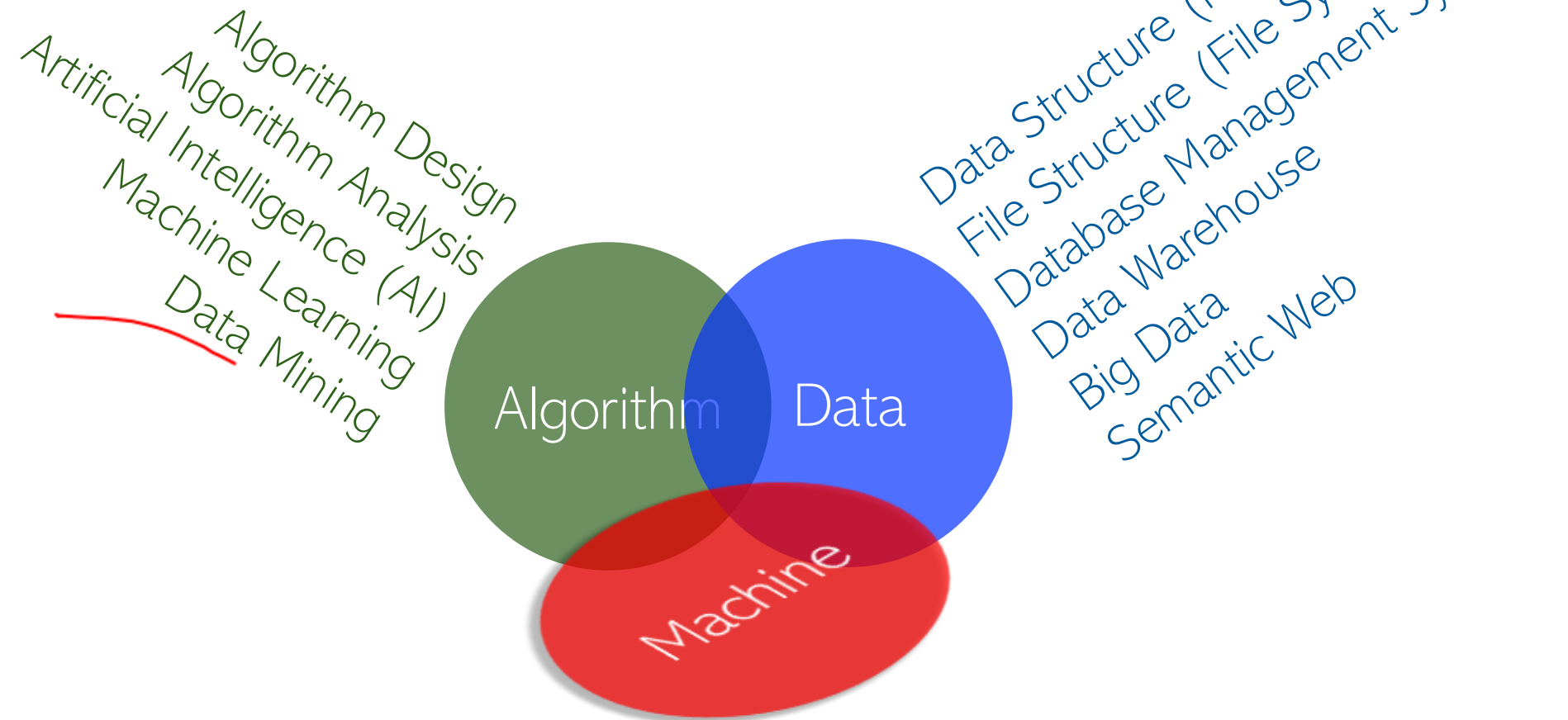
$$\text{—} + \text{●●●} = \text{●●●}$$

$$\text{●●●} - \text{—} = \text{●●●}$$



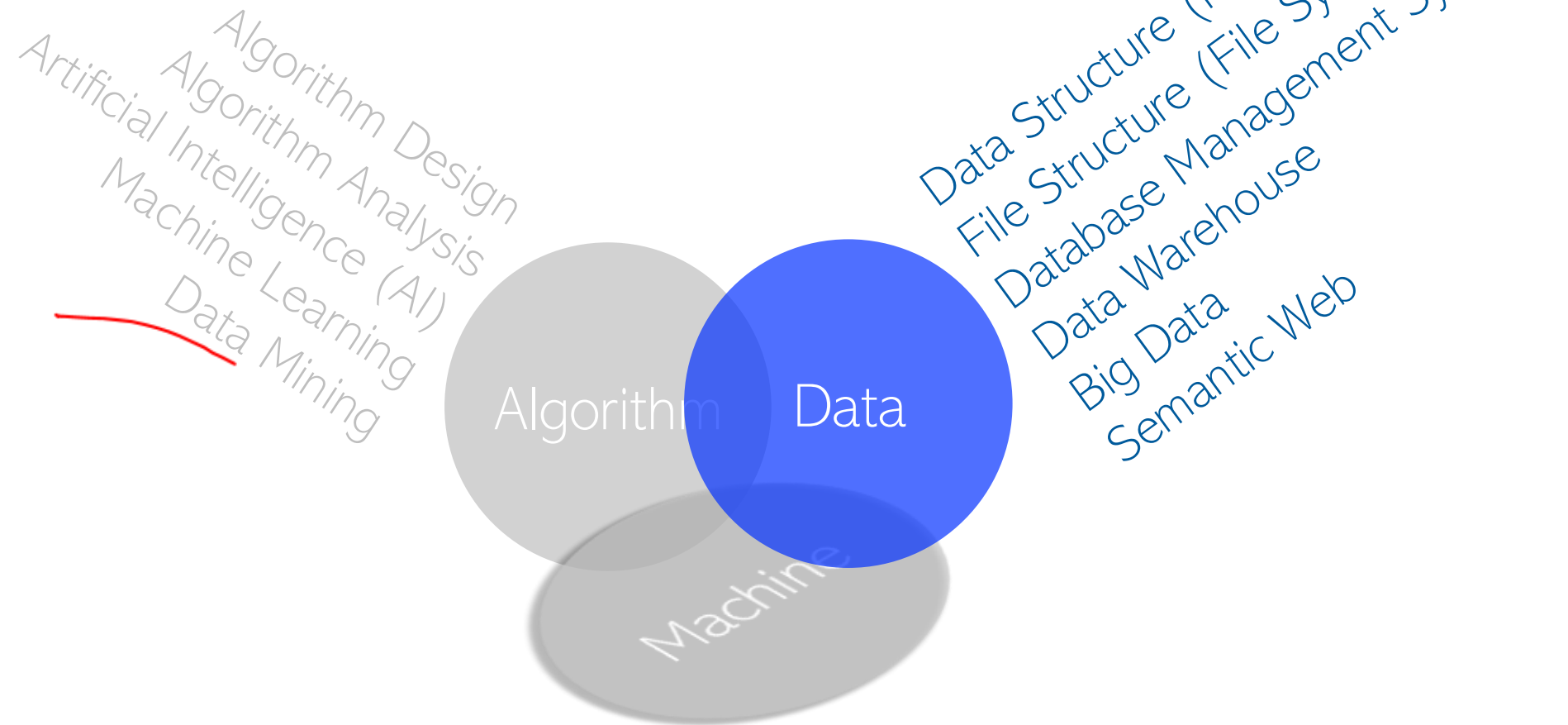
9

A Data Odyssey



Digital Design (Logic Circuits, qbits)
Computer Architecture (RISC, CISC, CPU, GPU, TPU, QC)
Programming Languages (Assembly, C, Java, Python, ...)
Operating Systems (Unix, Linux, ...)

A Data Mining Odyssey



Digital Design (Logic Circuits, qbits)
Computer Architecture (RISC, CISC, CPU, GPU, TPU, QC)
Programming Languages (Assembly, C, Java, Python, ...)
Operating Systems (Unix, Linux, ...)

Outline: W01 B

I) An Odyssey

II) DIKW Pyramid

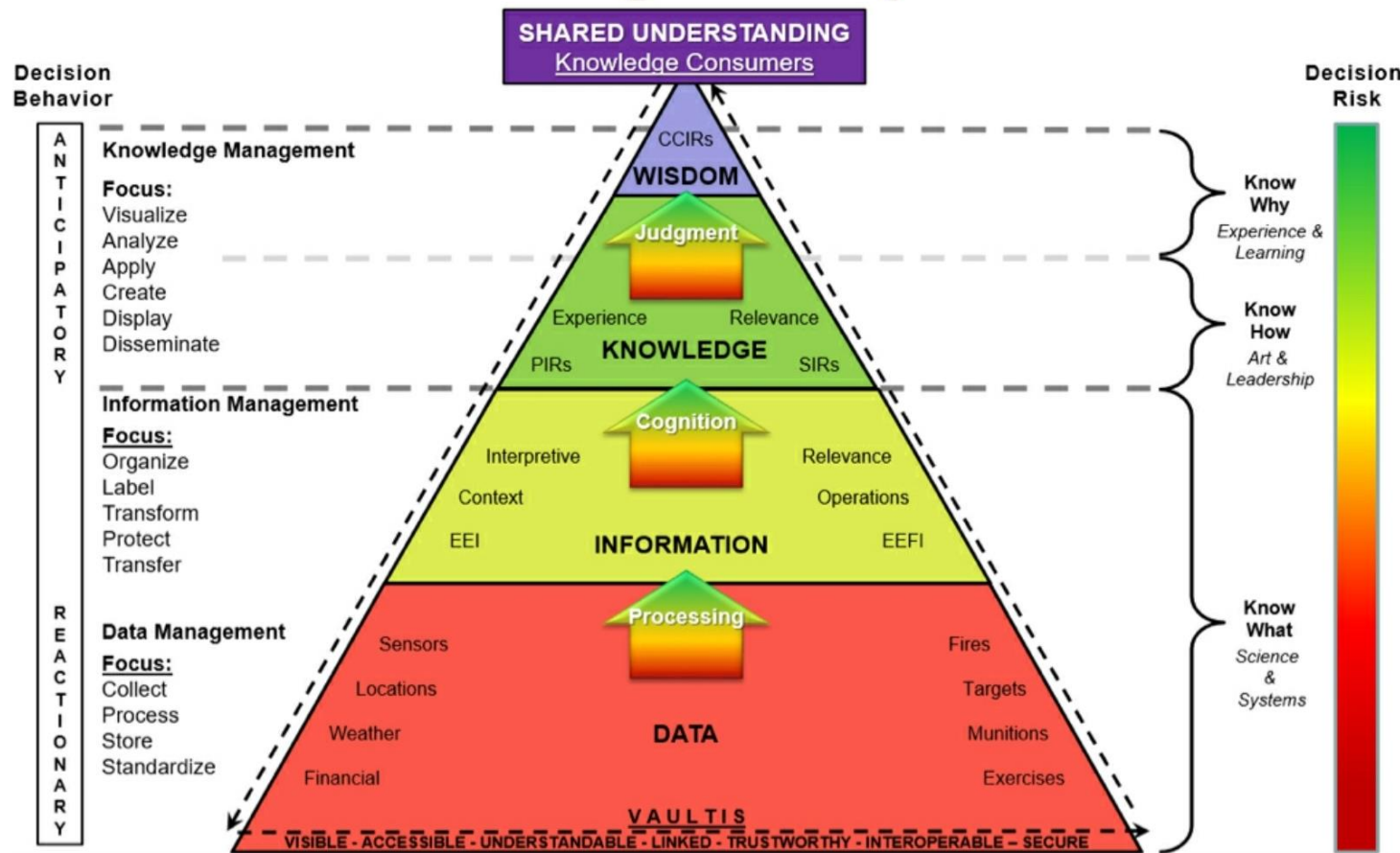
III) Data Model Abstractions

III) WWW to GGG

DIKW Pyramid

The DIKW Pyramid in the U.S. Department of Defense

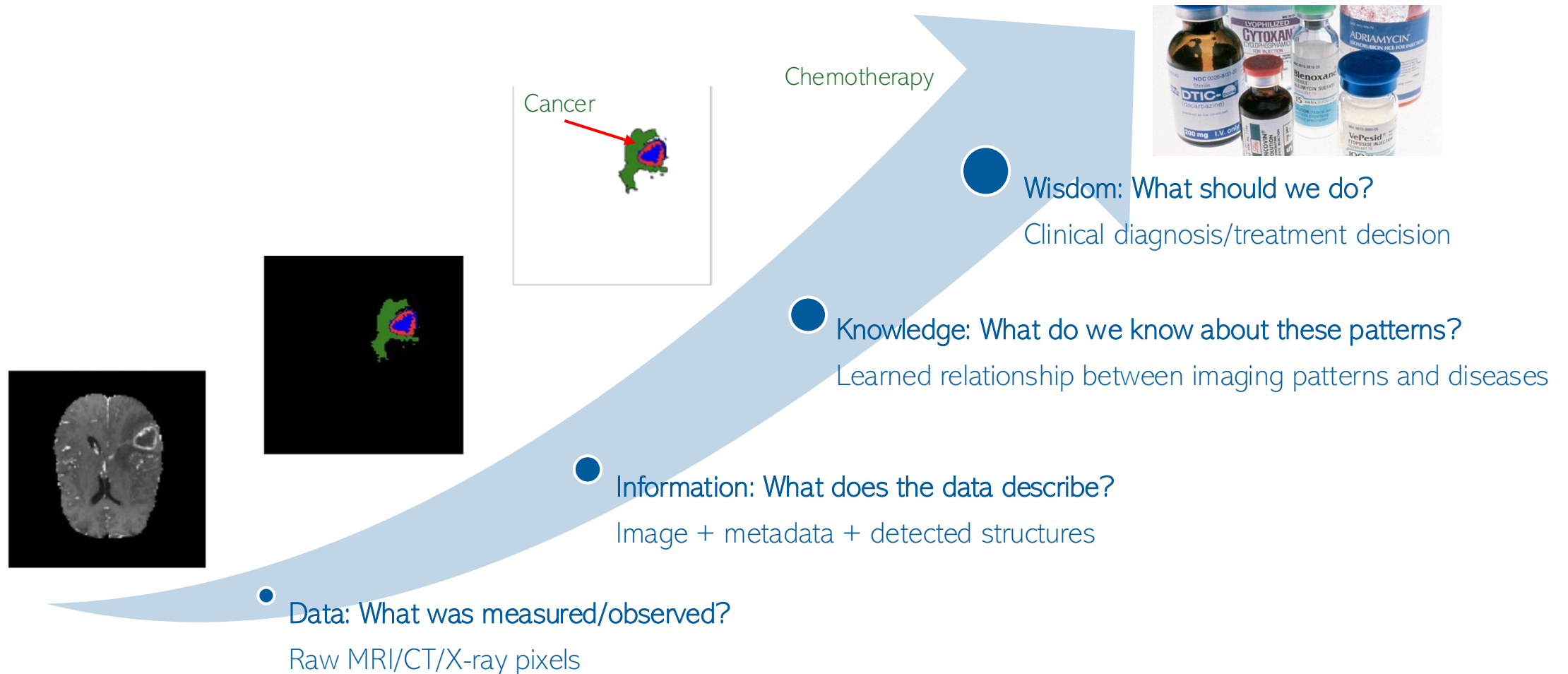
The KM Cognitive Pyramid



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DIKW Pyramid

Example in Biomedical Imaging



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Wisdom→Knowledge→Information→Data

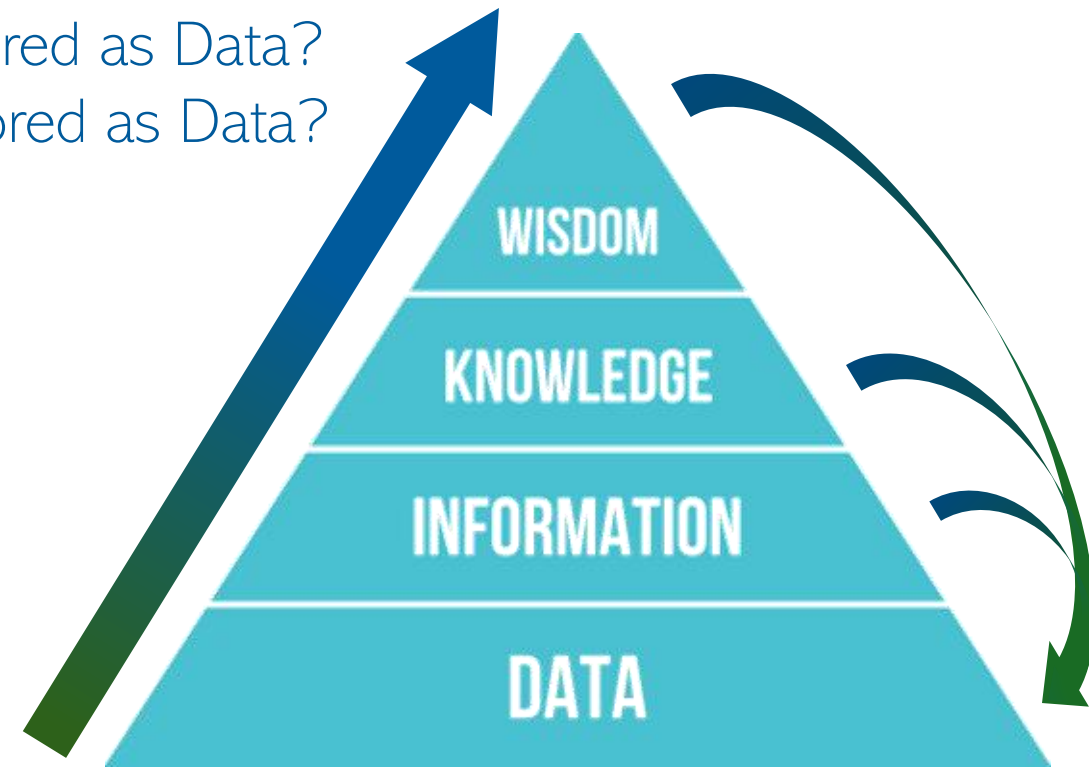
After the first button up, the result can be stored, but should it be stored like original raw data?

No!

What is an efficient model for **Wisdom** to be stored as Data?

What is an efficient model for **Knowledge** to be stored as Data?

What is an efficient model for **Information** to be stored as Data?



Outline: W01 B

I) An Odyssey

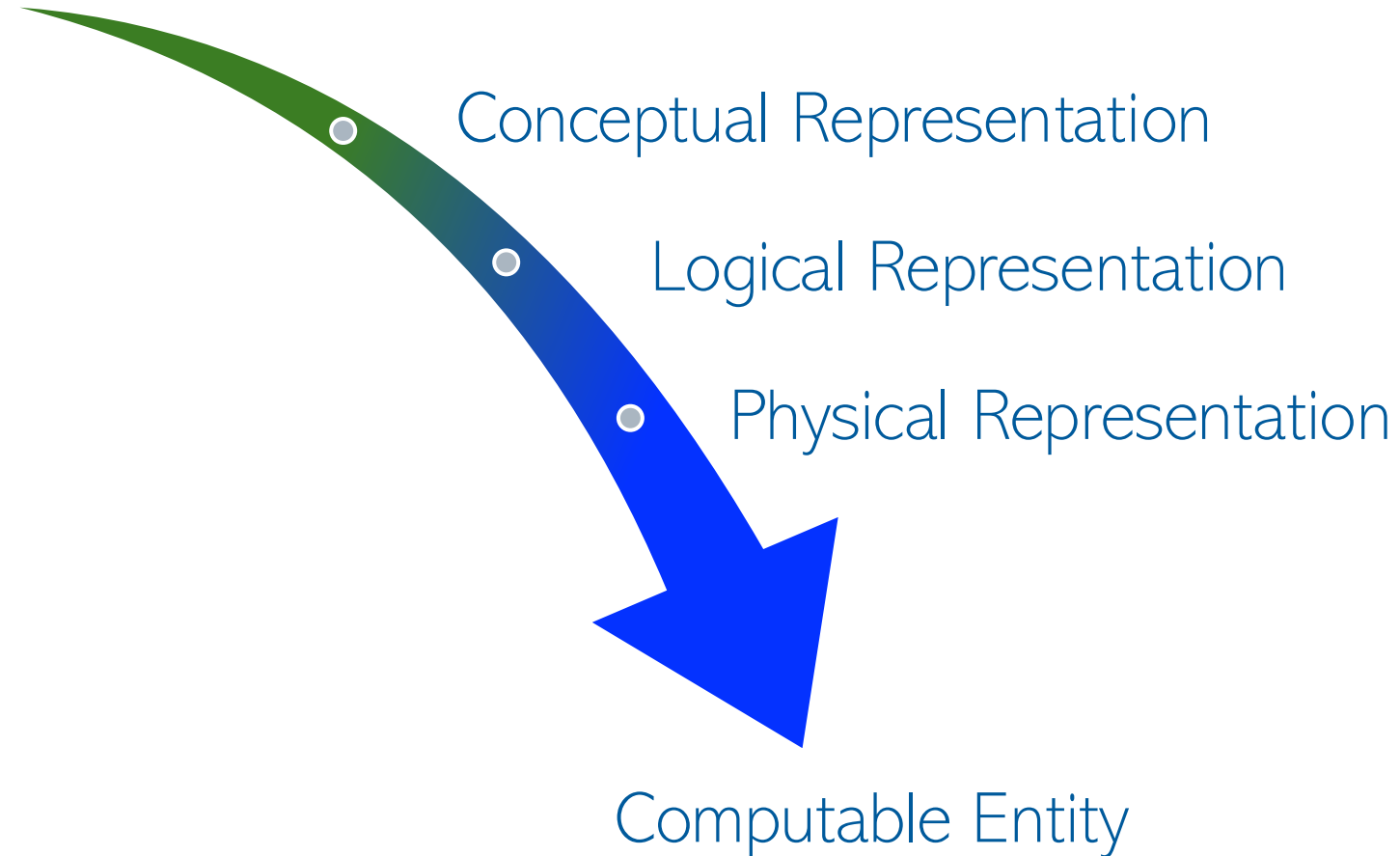
II) DIKW Pyramid

III) Data Model Abstractions

III) WWW to GGG

16 Data Model Abstractions

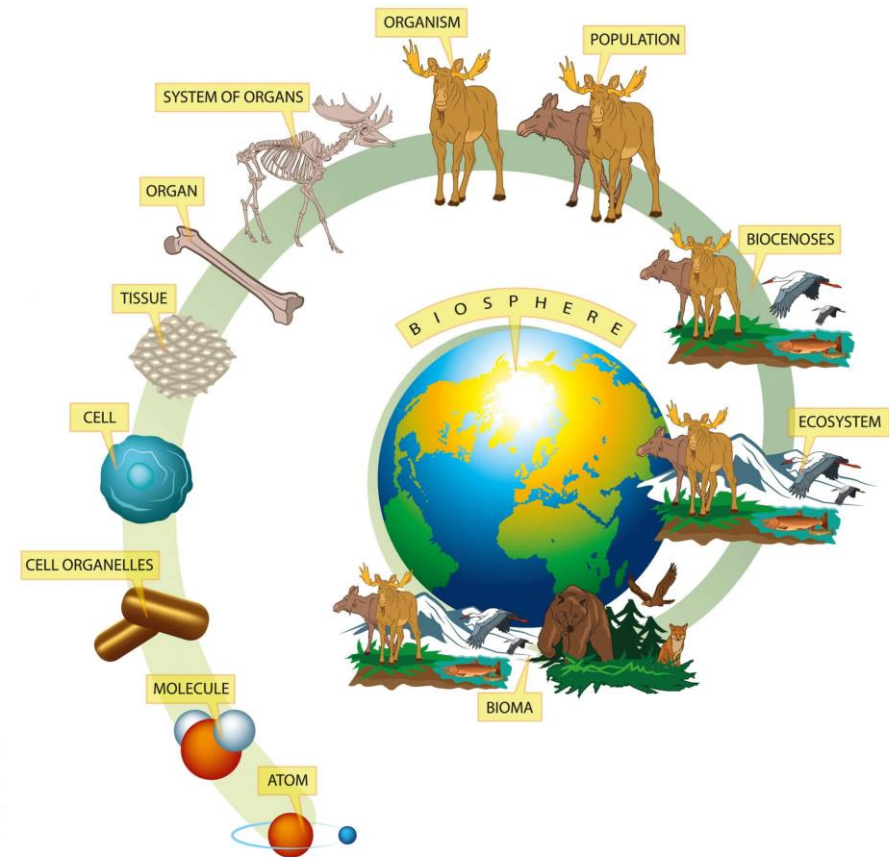
Real-World Observation



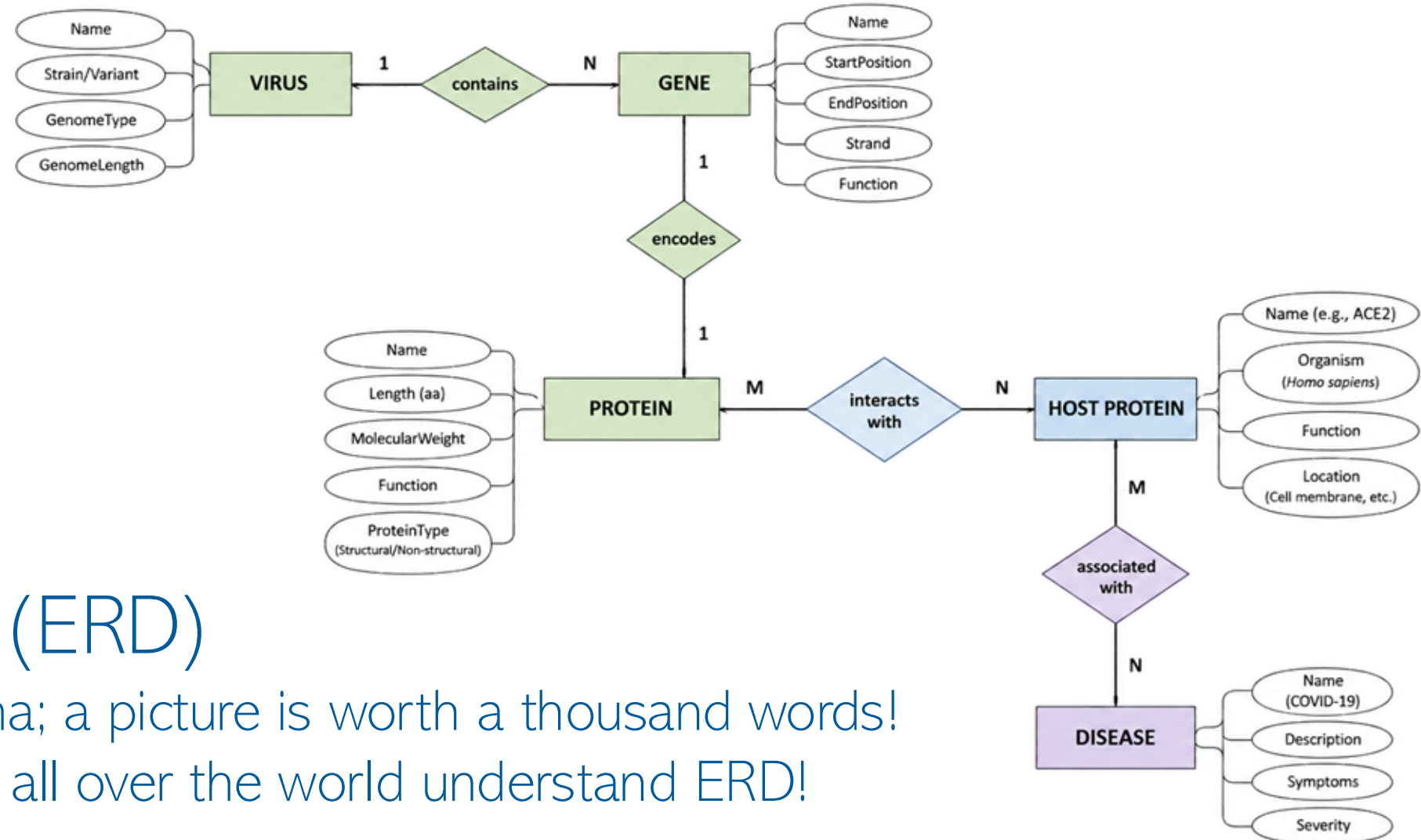
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Abstractions: Conceptual Representation

1. Identify Real-World Entities, Attributes, Relationships
2. Create Schema



Abstractions: Conceptual Representation

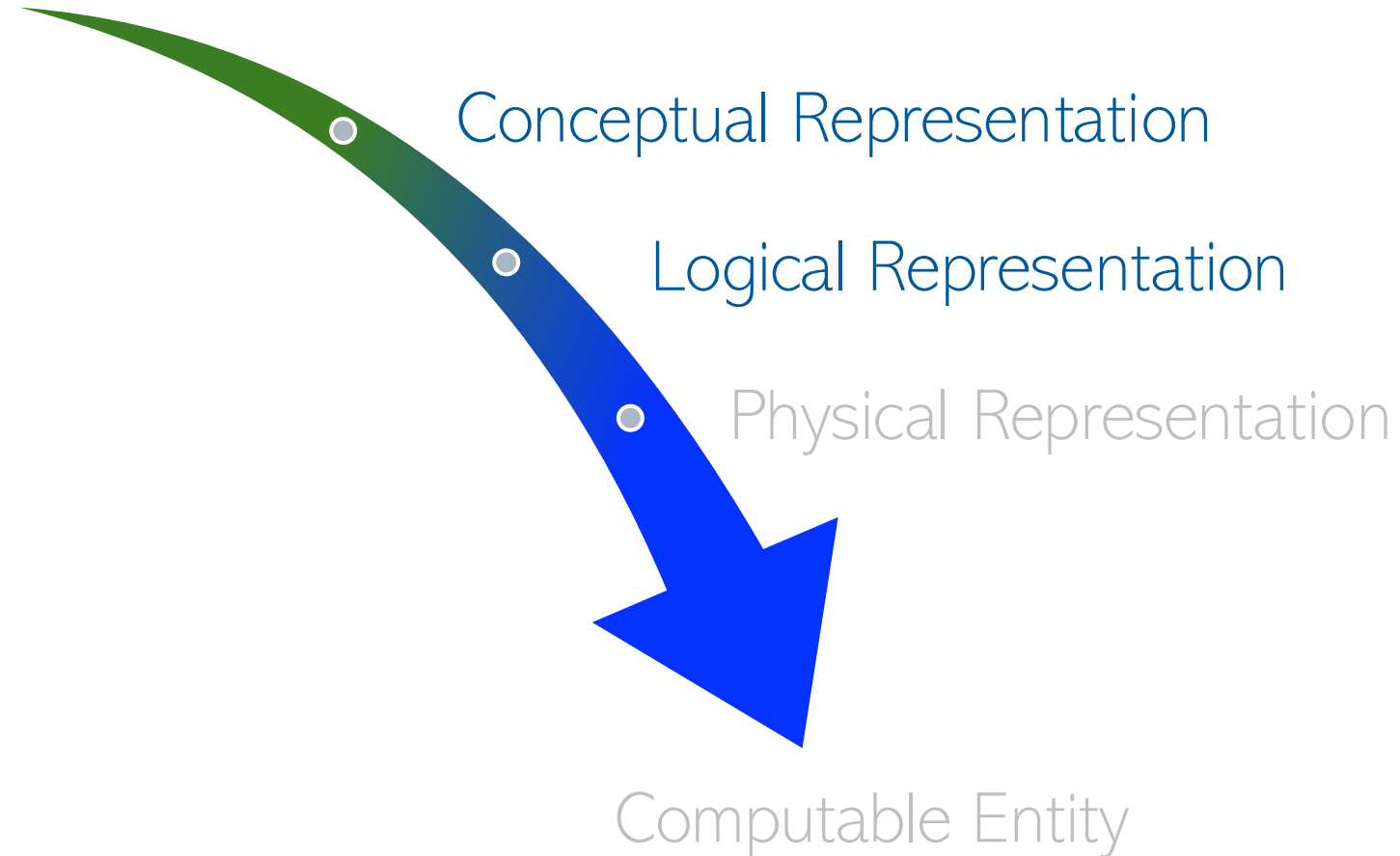


ER Diagram (ERD)

A graphical schema; a picture is worth a thousand words!
All data engineers all over the world understand ERD!

19 Data Model Abstractions

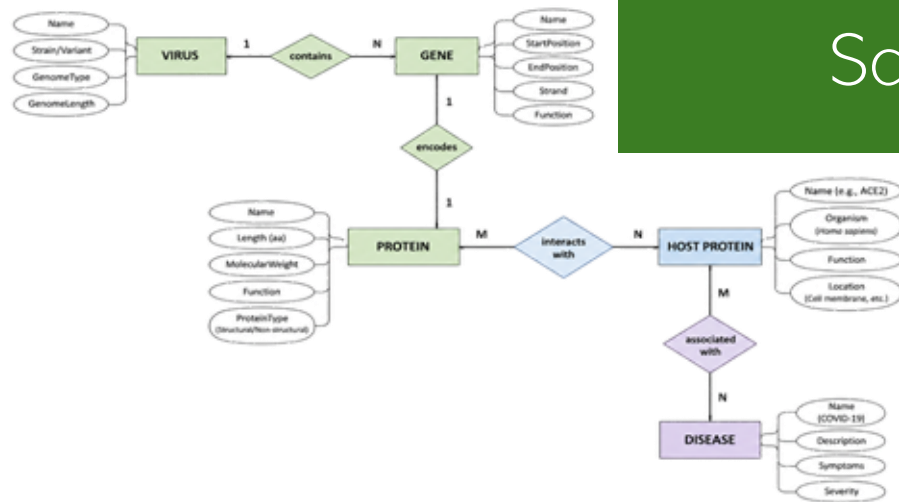
Real-World Observation



20

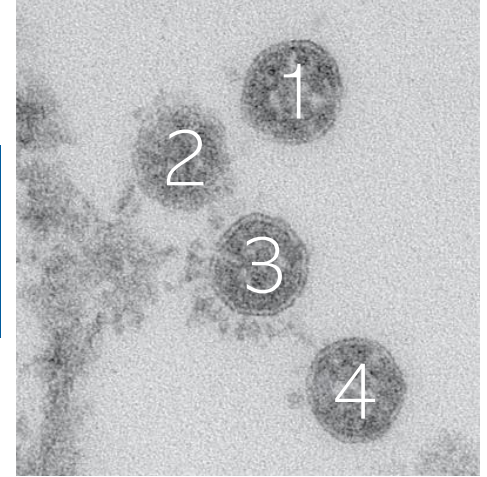
Abstractions: Logical Representation

How **instances** of entities, **values** of their attributes, their relationships should be represented based on a schema.



Schema

Real-World
Instances of Entity



Storage

Abstractions: Logical Representation

Data Schema

Data about Data: Meta-data

Defined at Setup Time

Rarely Change

Data Instance

Actual Data

Inserted at Running Time

Rapidly Change

Must Conform to Schema

Abstractions: Logical Representation

Tabular ...

Virus Entity				Gene Entity				
Name (string)	Variant (string)	Genome Type (string)	Genome Length (decimal)	Name (string)	Function (string)	Start (string)	End (decimal)	Strand (bool)
SARS-CoV-2	Wuhan-Hu-1	ssRNA	29,903 nt	ORF1ab	Polyprotein / replication	266	21,555	+
				ORF3a	Accessory protein	25,393	26,220	+
Influenza	A/Puerto Rico/8/1934 (H1N1)	ssRNA	13,588 nt	ORF1ab	Polyprotein / replication	266	21,555	+
				ORF3a	Accessory protein	25,393	26,220	+

Abstractions: Logical Representation

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				ORF3a	Accessory protein	25,393	26,220	+

Redundancy! Wasting space. Not that important ...

Abstractions: Logical Representation

Tabular ...

Virus Entity				Gene Entity				
Name (string)	Variant (string)	Genome Type (string)	Genome Length (decimal)	Name (string)	Function (string)	Start (string)	End (decimal)	Strand (bool)
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				ORF3a	Accessory protein	25,393	26,220	+

Inconsistency! Incoherent pieces of information.

Abstractions: Logical Representation

Tabular ...

Virus Entity				Gene Entity				
Name (string)	Variant (string)	Genome Type (string)	Genome Length (decimal)	Name (string)	Function (string)	Start (string)	End (decimal)	Strand (bool)
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				ORF3a	Accessory	25,393	26,220	+

Speed! Cascading of updates/deletions.

Abstractions: Logical Representation

Not All (Data) Representations are Efficient.

Not All Conceptual Representations are Efficient.

Not All Logical Representations are Efficient.

Not All Physical Representations are Efficient.

Variety of representations have been proposed ...

1960: Object Oriented (Course: OOP using Java/C++/...)

1969: Relational: Mathematical Relationships (Course: DBMS)

1996: XML (This Course)

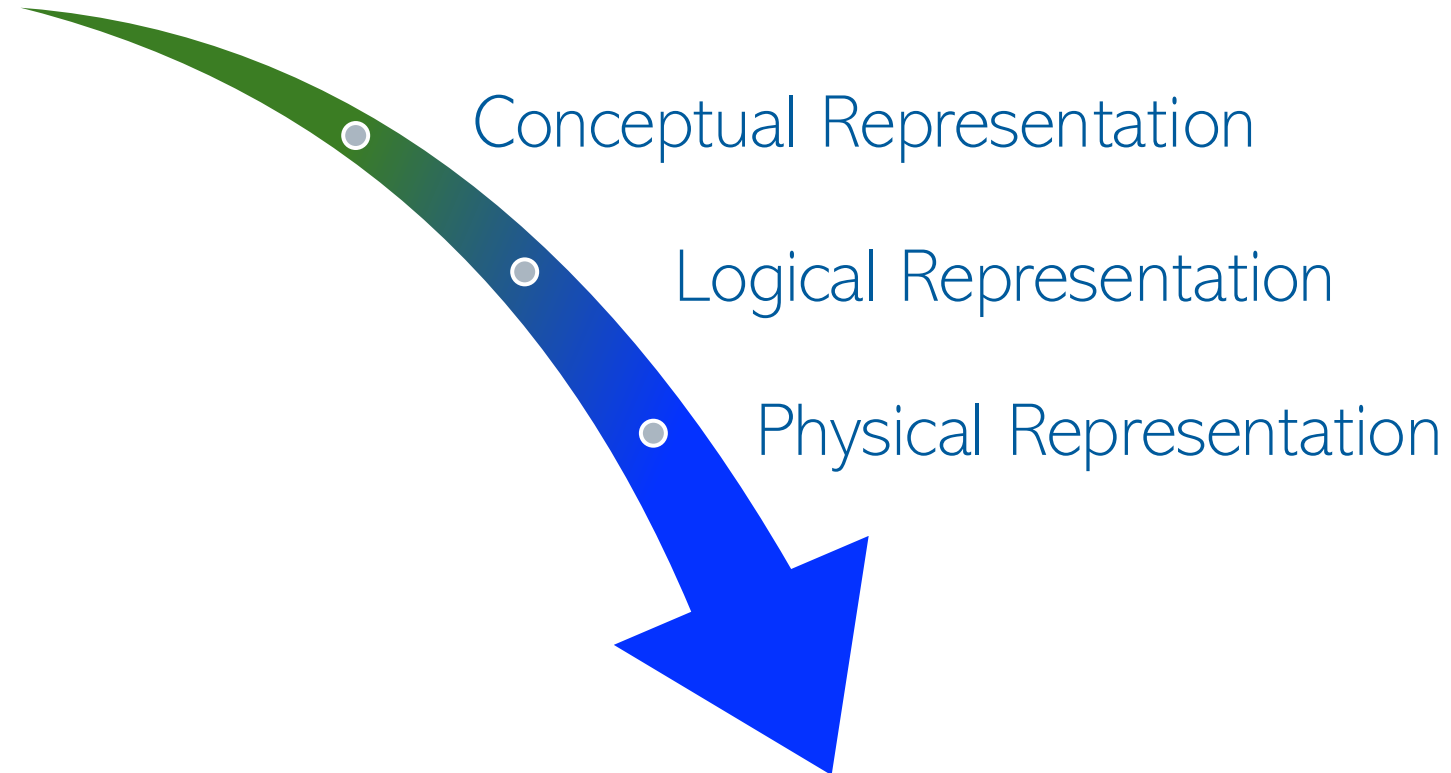
...

2007: Knowledge Graphs (This Course)

...

27 Data Model Abstractions

Real-World Observation



Computable Entity

Abstractions: Physical Representation

Where and how are the instances stored in **physical storages**.

Internet → Cloud → Database → File



Abstractions: DataBase Management Systems (DBMS)

Company	DBMS
Oracle	Oracle, MySQL
Microsoft	MySQLServer
IBM	DB2
Apache Software Foundation	CouchDB (NOSQL)

30 Outline: W01 B → W02A

I) An Odyssey

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III) WWW to GGG

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Sir Timothy John Berners-Lee (1955-Present), aka TimBL

English Computer Scientist, Professorial Research Fellow at the University of Oxford, Professor Emeritus at the MIT

1989: World Wide Web (WWW), HTML, URL, HTTP

For human-readable publicly shared

Information

Knowledge

But proposed Giant Global Graph (GGG)

For machine-readable publicly shared

Information

Knowledge



Online Encyclopedia → Wikipedia

Encyclopedia as the Fundamental Knowledge Source (Base)

Wikipedia as the **Online** Encyclopedia

So, Wikipedia as the Fundamental **Online** Knowledge Source (Base)

Not Machine-Readable

Not Machine-Understandable



Online Encyclopedia → Wikipedia

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Human: It's a thing in real world, but what is it?

Online Encyclopedia → Wikipedia

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Human:

It's a thing in real world

It's a movie

Made by Francis Ford Coppola

Actors are Brando and Pacino

Released in 1972

...

Understanding the semantic (meaning) is:
Correct mapping to real world object/concept

The Godfather

100 languages

Article Talk

Read Edit View history

From Wikipedia, the free encyclopedia

This article is about the 1972 film. For the original novel on which the film is based, see [The Godfather \(novel\)](#). For other uses, see [Godfather](#).

The Godfather is a 1972 American epic [gangster film](#)^{[2][3][4]} directed by [Francis Ford Coppola](#), who co-wrote the screenplay with [Mario Puzo](#) based on Puzo's best-selling 1969 novel. The film features an [ensemble cast](#) that includes [Marlon Brando](#), [Al Pacino](#), [James Caan](#), [Richard Castellano](#), [Robert Duvall](#), [Sterling Hayden](#), [John Marley](#), [Richard Conte](#) and [Diane Keaton](#). It is the first installment in *The Godfather trilogy*, which chronicles the [Corleone family](#) under patriarch [Vito Corleone](#) (Brando) and the transformation of his youngest son, [Michael Corleone](#) (Pacino), from reluctant family outsider to ruthless [Mafia boss](#).

[Paramount Pictures](#) obtained the rights to the novel for \$80,000, before it gained popularity.^{[5][6]} Studio executives had trouble finding a director; the first few candidates turned down the position before Coppola signed on to direct the film, but disagreement followed over casting several characters, in particular Vito (Brando) and Michael (Pacino). Filming took place primarily in locations around New York City and [Sicily](#), and it was completed ahead of schedule. The score was composed principally by [Nino Rota](#), with additional pieces by [Carmine Coppola](#).

The Godfather premiered at the [Loew's State Theatre](#) on March 14, 1972, and was widely released in the United States on March 24, 1972. It was the [highest-grossing film of 1972](#), and was for a time the [highest-grossing film ever made](#), earning between \$250 and \$291 million at the box office. The film was acclaimed by critics and audiences, who praised its performances—particularly those of Brando and Pacino—direction, screenplay, story, cinematography, editing, score, and [portrayal of the Mafia](#). *The Godfather* launched the successful careers of Coppola, Pacino, and other relative newcomers in the cast and crew. At the [45th Academy Awards](#), the film won [Best Picture](#), Brando won [Best Actor](#), and Puzo and Coppola won for [Best Adapted Screenplay](#). In addition, the seven other Oscar nominations included Pacino, Caan, and



Brando (right) and Pacino as Don Vito and Michael Corleone, respectively

Francis Ford Coppola
Based on [The Godfather](#) by Mario Puzo
Produced by [Albert S. Ruddy](#)
Starring [Marlon Brando](#)
[Al Pacino](#)
[James Caan](#)
[Richard Castellano](#)

Online Encyclopedia → Wikipedia

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Machine: ?

Online Encyclopedia → Wikipedia

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Machine: Emulate/Showing Correct Mapping to Real-World Object

```
ns0:thegodfather ns0:type Movie.  
ns0:thegodfather ns0:title "The Godfather".  
ns0:thegodfather ns0:releasedate "03/24/1972".  
ns0:thegodfather ns0:director <ns0:_x>.  
ns0:_x ns0:name "Francis Ford Coppola".  
ns0:_x ns0:imdbpage "www.imdb.com/name/nm0000338/"  
ns1:francis_coppola ns1:name "Francis Ford Coppola".  
ns1:francis_coppola ns1:birthday "04/07/1939".  
ns1:francis_coppola ns1:birthplace "Detroit".  
ns1:francis_coppola ns1:imdbpage "www.imdb.com/name/nm0000338/"
```

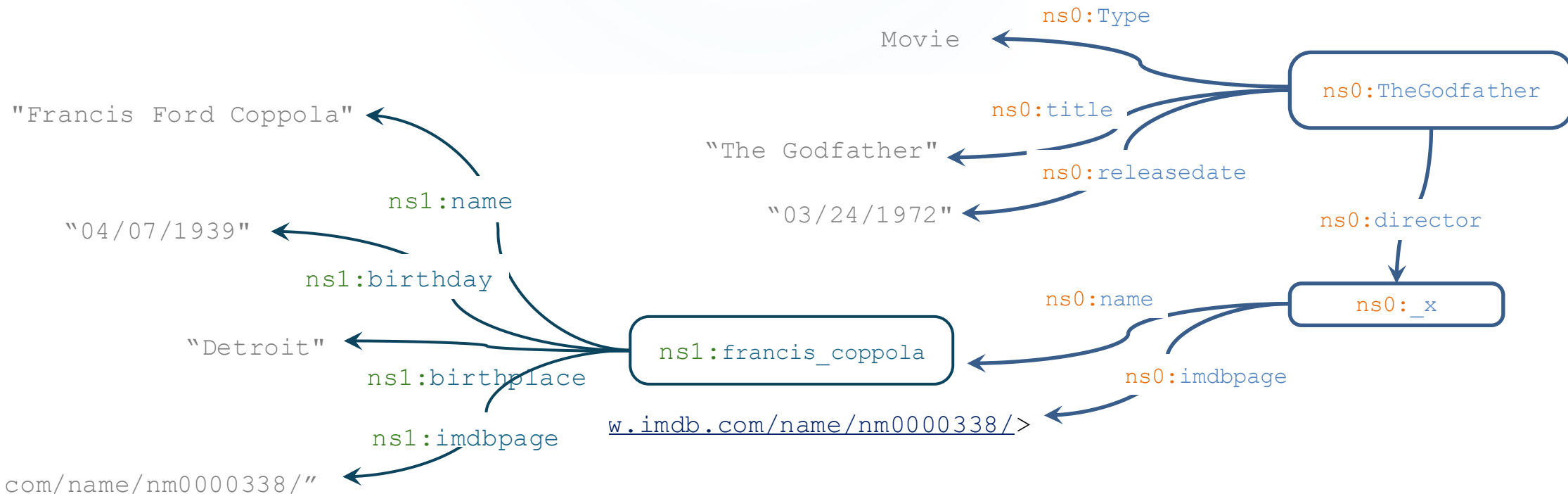
Online Encyclopedia → Wikipedia

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Machine: Triples (Subject, Predicate, Object)



Online Encyclopedia → Wikipedia → DBpedia

World Wide Web (WWW): a web of discrete information objects readable by human beings

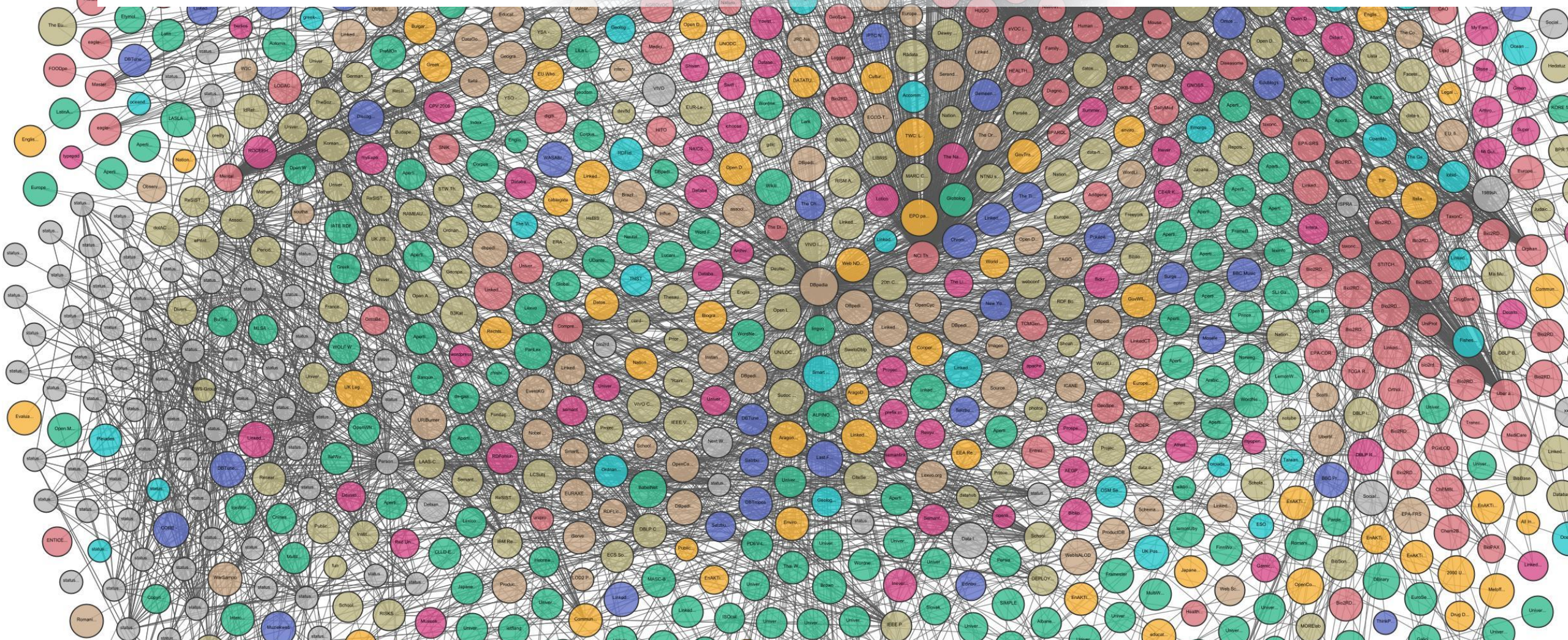
→ **Giant Global Graph (GGG)**: a web of information objects and their relationships that conceptually or semantically link to each other for human- and machine-created data objects

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2007: DBpedia Knowledge Graph (KG)

2021: Over 850 million semantic triples

<https://lod-cloud.net/>



2007: DBpedia Knowledge Graph (KG)

https://en.wikipedia.org/wiki/Tim_Berners-Lee

https://dbpedia.org/page/Tim_Berners-Lee

Tim Berners-Lee

Article Talk

Read View source View history

From Wikipedia, the free encyclopedia

Sir Timothy John Berners-Lee (born 8 June 1955),^[1] also known as **TimBL**, is an English **computer scientist** best known as the inventor of the **World Wide Web**, **HTML**, the **URL** system, and **HTTP**. He is a professorial research fellow at the **University of Oxford**^[2] and a professor **emeritus** at the **Massachusetts Institute of Technology** (MIT).^{[3][4]}

Berners-Lee proposed an information management system on 12 March 1989^{[5][6]} and implemented the first successful communication between a Hypertext Transfer Protocol (HTTP) client and **server** via the **Internet** in mid-November.^{[7][8][9][10][11]} He devised and implemented the first Web browser and Web server and helped foster the Web's subsequent development. He is the founder and emeritus director of the **World Wide Web Consortium** (W3C), which oversees the continued development of the Web. He co-founded (with **Rosemary Leith**) the **World Wide Web Foundation**. In 2009, he was elected **Foreign Associate of the National Academy of Sciences**.^{[12][13]}

Berners-Lee was previously a senior researcher and holder of the **3Com** founder's chair at the **MIT Computer Science and Artificial Intelligence Laboratory** (CSAIL).^[14] He is a director of the **Web Science Research Initiative** (WSRI)^[15] and a member of the advisory board of the **MIT Center for Collective Intelligence**.^{[16][17]} In 2011, he was named a member of the board of trustees of the **Ford Foundation**.^[18] He is a founder and president of the **Open Data Institute** and is an advisor at social network **MeWe**.^[19] In 2004, Berners-Lee was knighted by Queen **Elizabeth II** for his pioneering work.^{[20][21]} He received the 2016 **Turing Award** "for inventing the World Wide Web, the first web browser, and the fundamental protocols and algorithms allowing the Web to scale".^[22] He was named in *Time* magazine's list of the **100 Most Important People of the 20th century** and has received a **number of other accolades** for his invention.^[23]

Sir Tim Berners-Lee
OM KBE FRS RDI FRSA DFBCS FREng



Berners-Lee in 2025

Born	Timothy John Berners-Lee 8 June 1955 (age 71) London, England
Other names	TimBL · TBL
Education	University of Oxford (BA)
Known for	Invention of the World Wide Web , HTML
Spouses	Jane Northcote (m. 1976, divorced) Nancy Carlson (m. 1990; div. 2011) Rosemary Leith (m. 2014)
Children	2 children 3 step-children
Parents	Conway Berners-Lee (father) Mary Lee Woods (mother)
Awards	Turing Award (2016)



Browse using Formats

Faceted Browser Sparql Endpoint

About: [Tim Berners-Lee](#)

An Entity of Type: [animal](#), from Named Graph: <http://dbpedia.org>, within Data Space: [dbpedia.org](#)

English computer scientist, inventor of the World Wide Web (born 1955)



Property	Value
dbo:alias	TBL ^(en) TimBL ^(en)
dbo:birthDate	1955-06-08 (xsd:date)
dbo:birthName	Timothy John Berners-Lee ^(en)
dbo:birthPlace	dbr:London
dbo:birthYear	1955-01-01 (xsd:year)
dbo:description	English computer scientist, inventor of the World Wide Web (born 1955) ^(en)
dbo:education	dbr:The Queen's College, Oxford
dbo:knownFor	dbr:World Wide Web
dbo:spouse	dbr:Rosemary Leith
dbo:thumbnail	wiki-commons:Special:FilePath/L33_4919_(cropped).jpg?width=300

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2012: Google Knowledge Graph: things, not strings

<https://www.youtube.com/watch?v=mmQl6VGvX-c>



Introducing the Knowledge Graph



Subscribe

9.4K



Share

Ask

Save



1.6M views 14 years ago

The Knowledge Graph is a huge collection of the people, places and things in the world and how they're connected to one another. With this Search technology, Google can get you the best possible answers and help jump start your discovery. Get an under the hood look at the next frontier, from the team at Google behind the technology.

...more

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2023: SPOKE

The Scalable Precision medicine Open Knowledge Engine

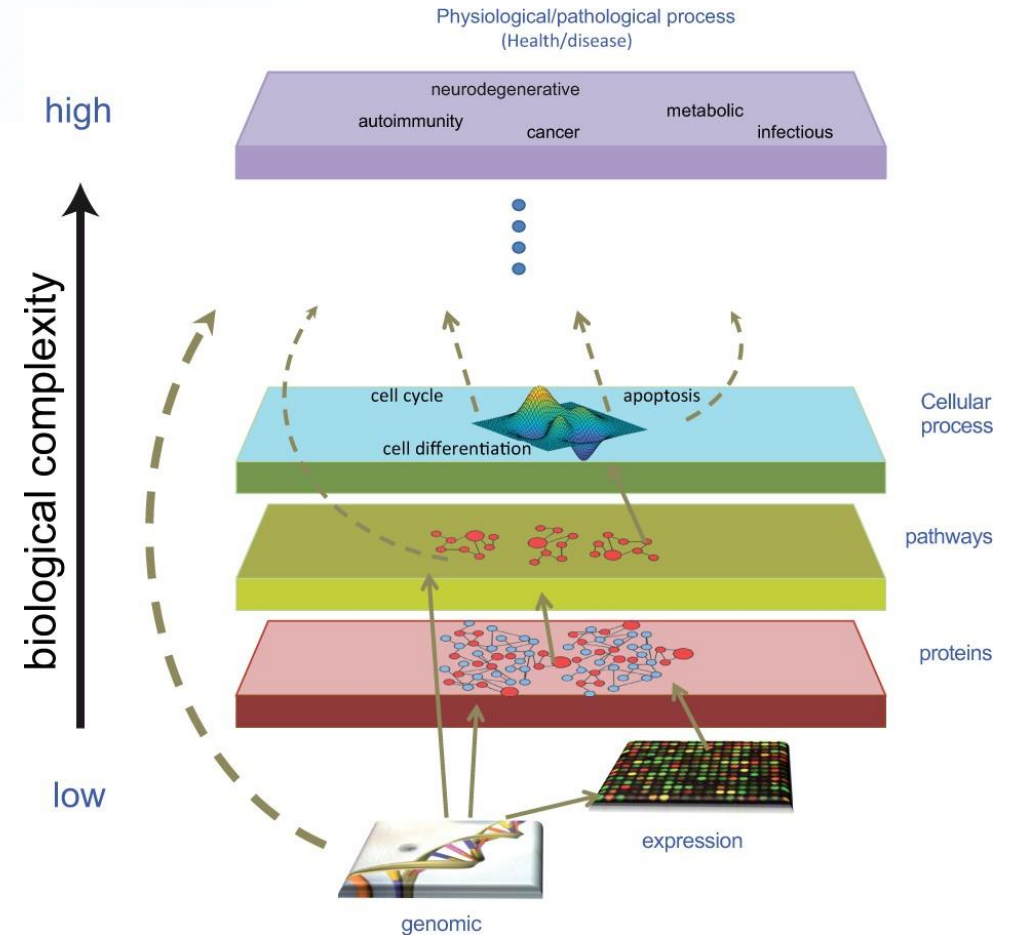
A massive knowledge graph of biomedical information

<https://doi.org/10.1093/bioinformatics/btad080>

<https://spoke.rbvi.ucsf.edu/>

The screenshot shows the SPOKE web interface. At the top, there's a navigation bar with various folders like 'ri', 'Art', 'Data Visualization', 'Python', 'ML', 'UWindsor', 'NLP', 'IR', 'StarCSVenues', 'Nserc', 'COMP2650', 'UndergradResear...', 'TeamFormation', and '2Read'. Below this is the SPOKE logo and a search bar with a dropdown menu set to 'Anatomy (name)'. There are buttons for 'Submit', 'Find in network', 'Options', 'Sample Queries', and 'Legend'. Under the 'Sample Queries' section, there is a list of queries with their corresponding SMILES strings:

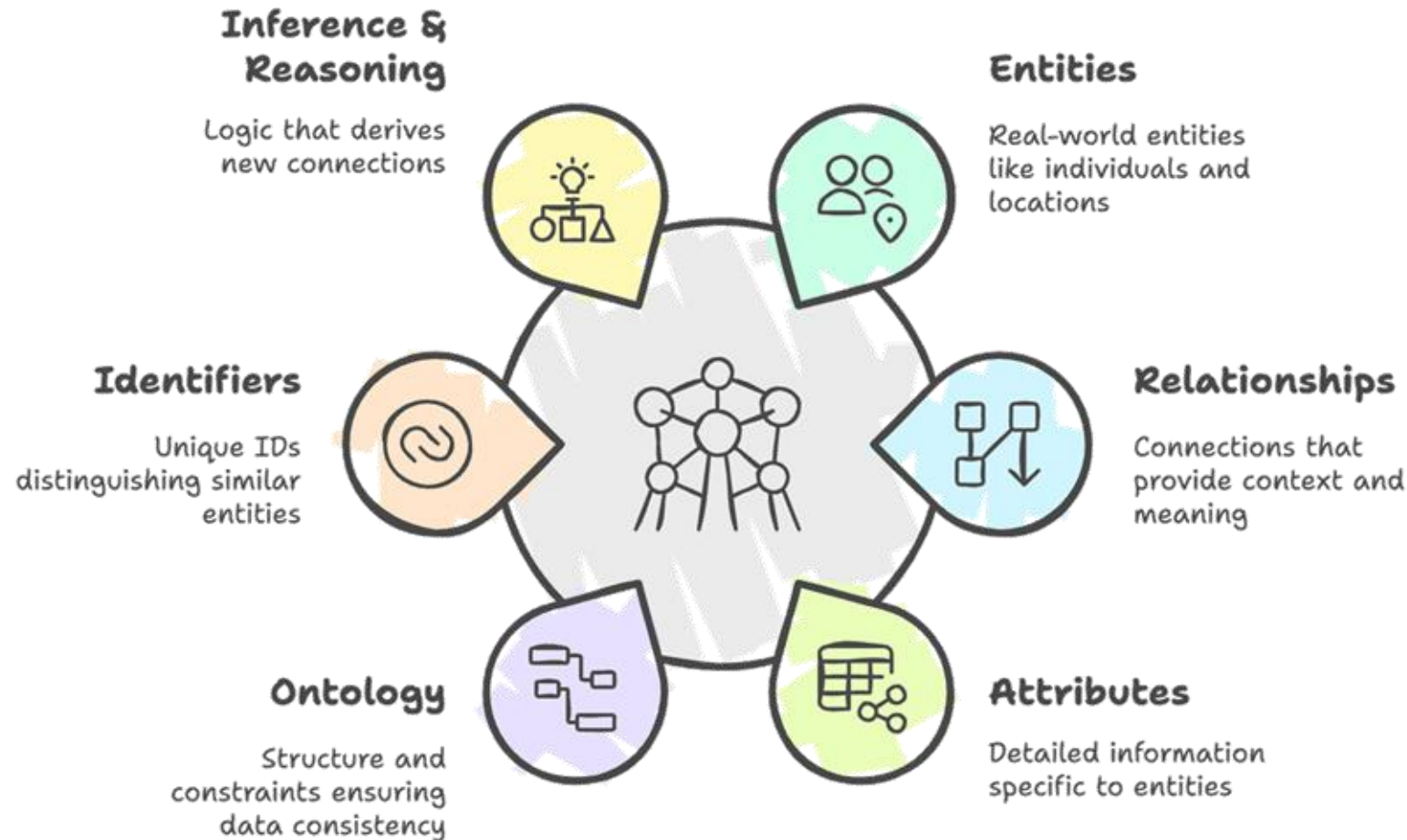
- Thyrotropin receptor (TSHR_HUMAN)
- Parathyroid hormone receptor (PTH1R_HUMAN)
- Ryanodine receptor 1 (RYR1_HUMAN)
- Serine/threonine-protein kinase B-raf (BRAF_HUMAN)
- Cystic fibrosis transmembrane conductance regulator (CFTR_HUMAN)
- Procaine (SEA search) (CCN(CC)CCOC(=O)c1ccc(N)cc1)
- Metformin (SEA search) (CN(C)C(=N)NC(=N)N)
- Chloroquine (SEA search) (CCN(CC)CCCC(C)Nc1ccnc2cc(Cl)ccc12)



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Knowledge Graph (KG) + Other Components

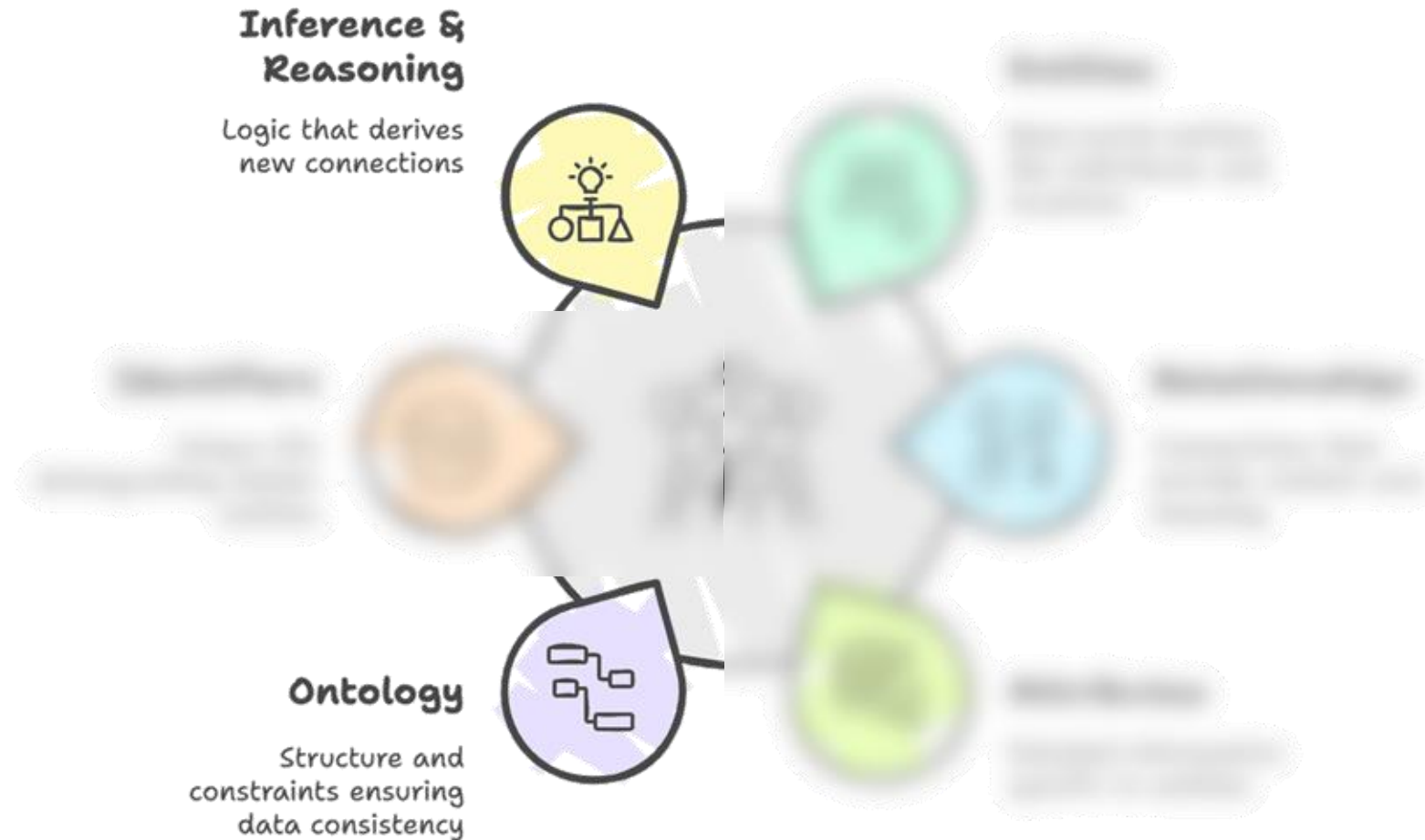
It's not just a graph!



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Knowledge Graph (KG)

+ Semantic Web: Ontology & Reasoning



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Knowledge Graph (KG)

+ Semantic Web: Ontology & Reasoning

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Human: What's the birthplace of The Godfather's director?

Director is a **person** → Francis Ford Coppola

Birthplace is a **city** → Francis Ford Coppola's birthplace → **Detroit**

You may right SQL query with subqueries/joins/...

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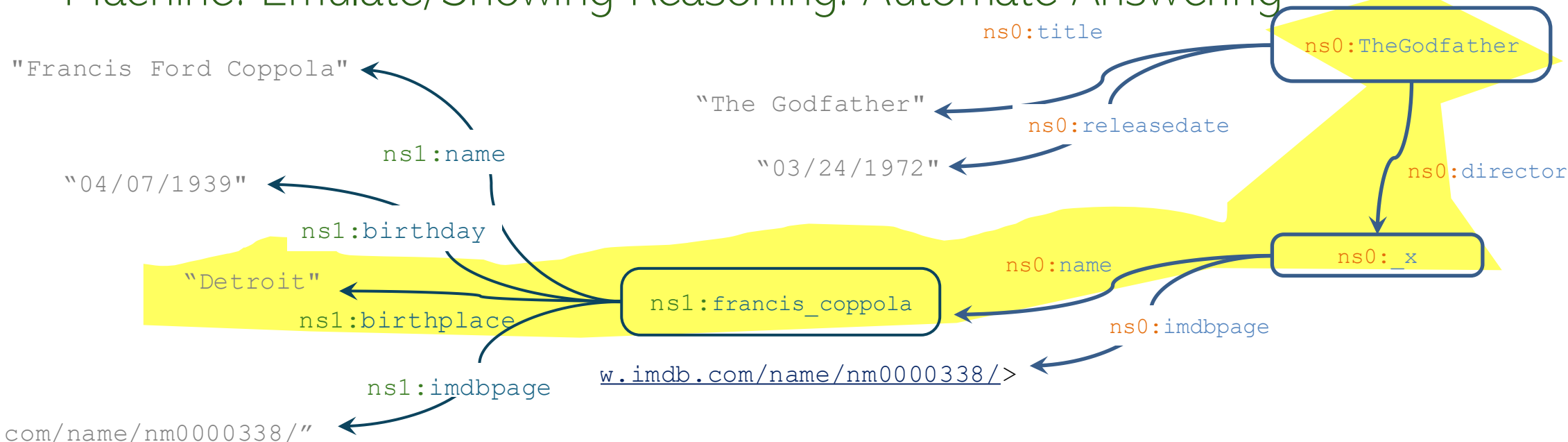
Knowledge Graph (KG) + Semantic Web: Ontology & Reasoning

Data: [T][h][e][][G][o][d][f][a][t][h][e][r]

Information: The Godfather

Knowledge:

Machine: Emulate/Showing Reasoning: Automate Answering



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A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

A Data Odyssey: DataBase Management Systems (DBMS)

Company	DBMS
Oracle	Oracle, MySQL
Microsoft	MySQLServer
IBM	DB2
Apache Software Foundation	CouchDB (NOSQL)
OpenLink Software	Virtuoso Universal Server Triple Store, SPARQL

Hyperlinks:

Constellation Fornax: <https://hubblesite.org/contents/media/videos/2014/27/766-Video.html>

Maya Civilization: https://en.wikipedia.org/wiki/Maya_numerals

Memento: <https://www.youtube.com/watch?v=ZABqyPb9HfA>

DIKW Size: <https://www.martinhilbert.net/worldinfocapacity-html/>
<https://pdodds.w3.uvm.edu/files/papers/others/2011/hilbert2011a.pdf>

Bruno Mars: <https://www.youtube.com/watch?v=LjhCEhWiKXk>

Google KG: <https://www.youtube.com/watch?v=mmQl6VGvX-c>

SPOKE KG: <https://spoke.rbvi.ucsf.edu/>